2.1 Introduction

The building material used for centuries was stone and wood, later bricks. During the 19th century, cast iron and steel were used. The 20th century brought concrete and reinforced concrete into civil engineering field. The end of 20th century had significant use of cement concrete composites, plastic, glass, new forms of steel, aluminum and insulation materials.

2.2 Cement Concrete Composites

2.2.1 Light weight concrete (LWC)

Light weight concrete is made by using light weight aggregates like foamed slag, bloated clay, sintered fly ash, rice husk, etc. It is also achieved by introducing larger voids using aluminum powder or by omitting the fine aggregate from the concrete mix.





(i) Advantages of LWC

1. Light weight concrete reduces the dead load of the structure.
2. It increases the progress of construction of the structure.
3. It lowers the haulage and handling charges.
4. If walls and floors are made of light weight concrete, the foundations will be lighter resulting in considerable economy in the construction.
5. The thermal conductivity of light weight concrete is relatively low, which prevents the heat transfer from roof and walls, resulting lower temperature inside the building.
6. In case of buildings where air conditioning is to be installed, the use of LWC has been found advantageous both in thermal comfort and low power consumption.

2.2.2 Autoclaved Aerated Concrete (AAC)

AAC is a lightweight precast form concrete building material. It is composed of quartz sand, calcined gypsum, lime, cement, water and aluminium power. Aluminum powder is used at a rate of 0.05%–0.08% by volume (depending on the pre-specified density). AAC products are cured under heat and pressure in an autoclave. The various forms of AAC are blocks, wall panels, floor and roof panels, cladding panels and lintels.

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